

LEAF-Net

A Hybrid Architecture Combining GBDT Leaf Embeddings
with Deep Cross-Attention Networks for Real Estate Valuation

Leaf Encoding + Embedding+ Attention-Fusion + Features Crossing

LEAF-Net

Basic

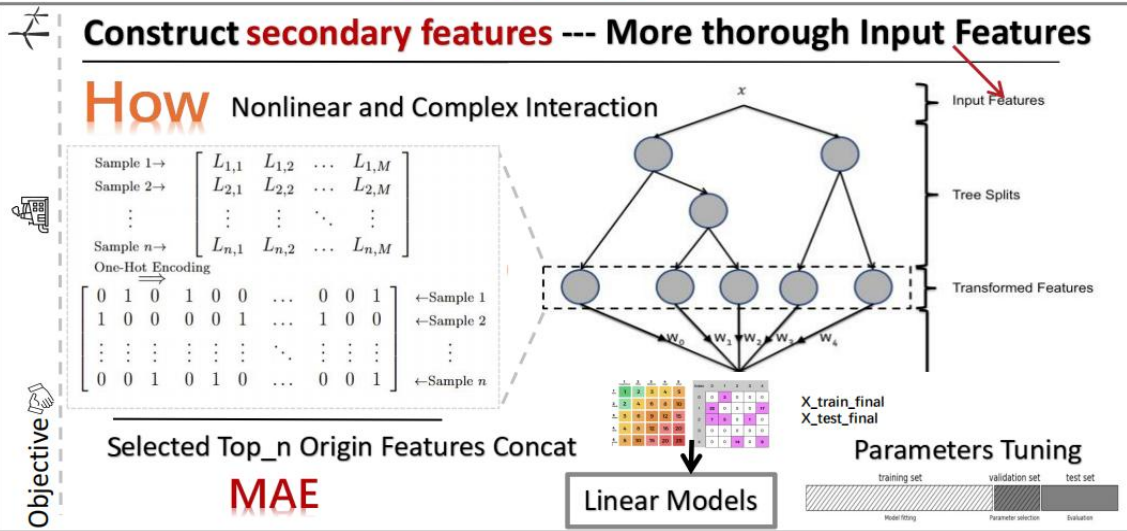
Key points should be carefully extracted from every column

High Data Quality



Excellent feature engineering

Leaf Encoding



Score in Kaggle

1 Calero Taydus



78.4

65

LEAF-Net Models

78.4

Singal DCN-v2

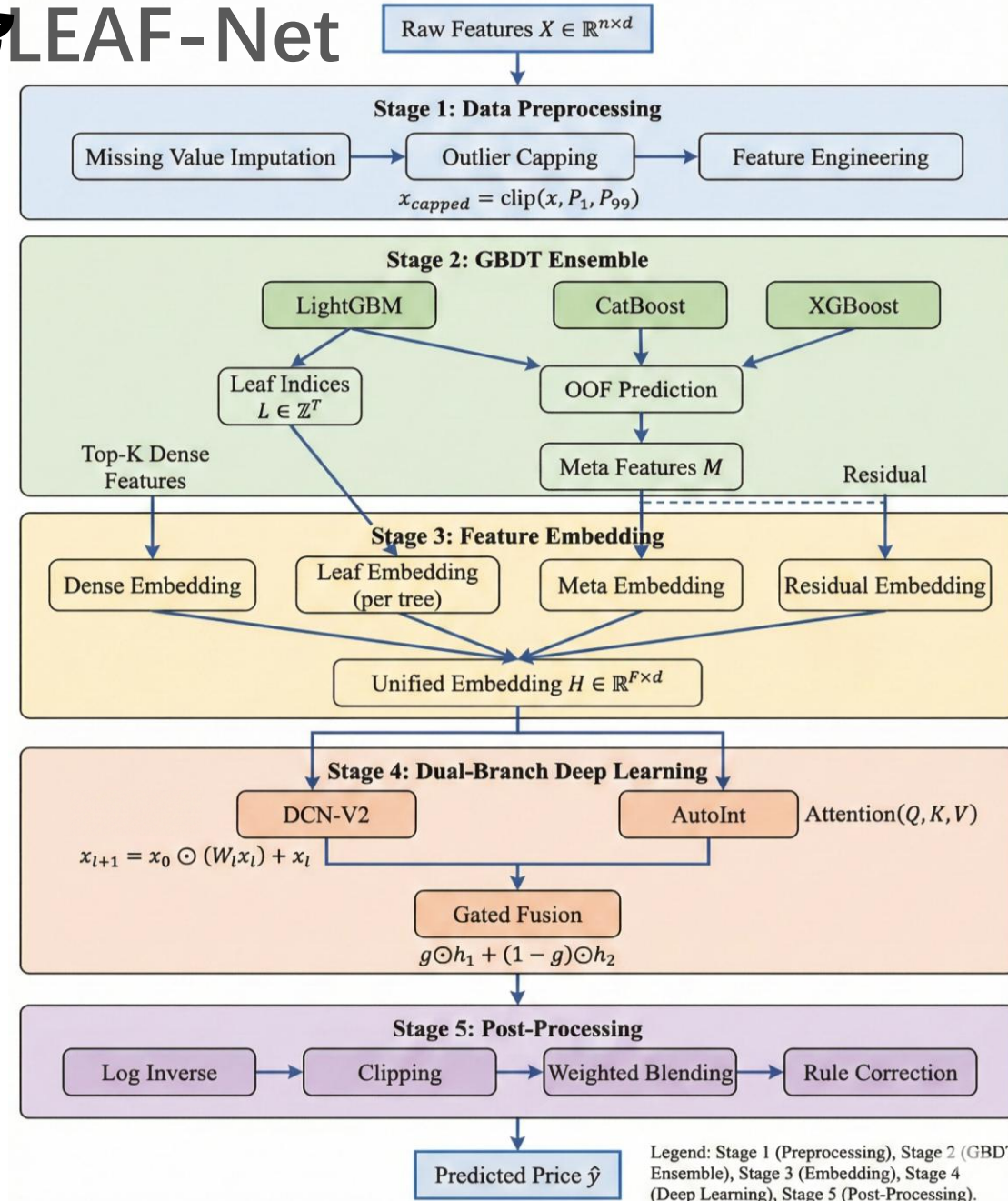
77.30

Best Lightgbm

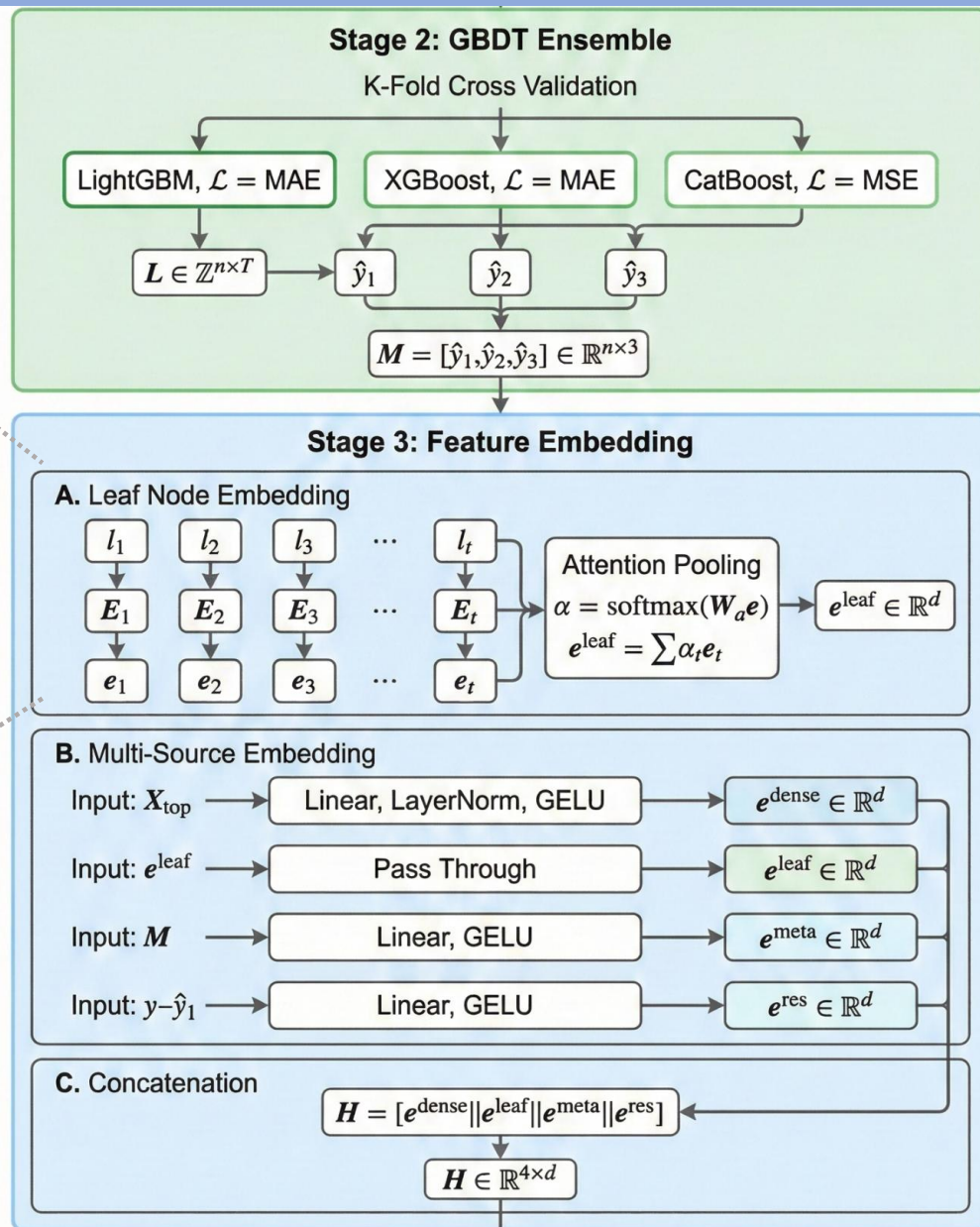
77.1

Best Linear Model

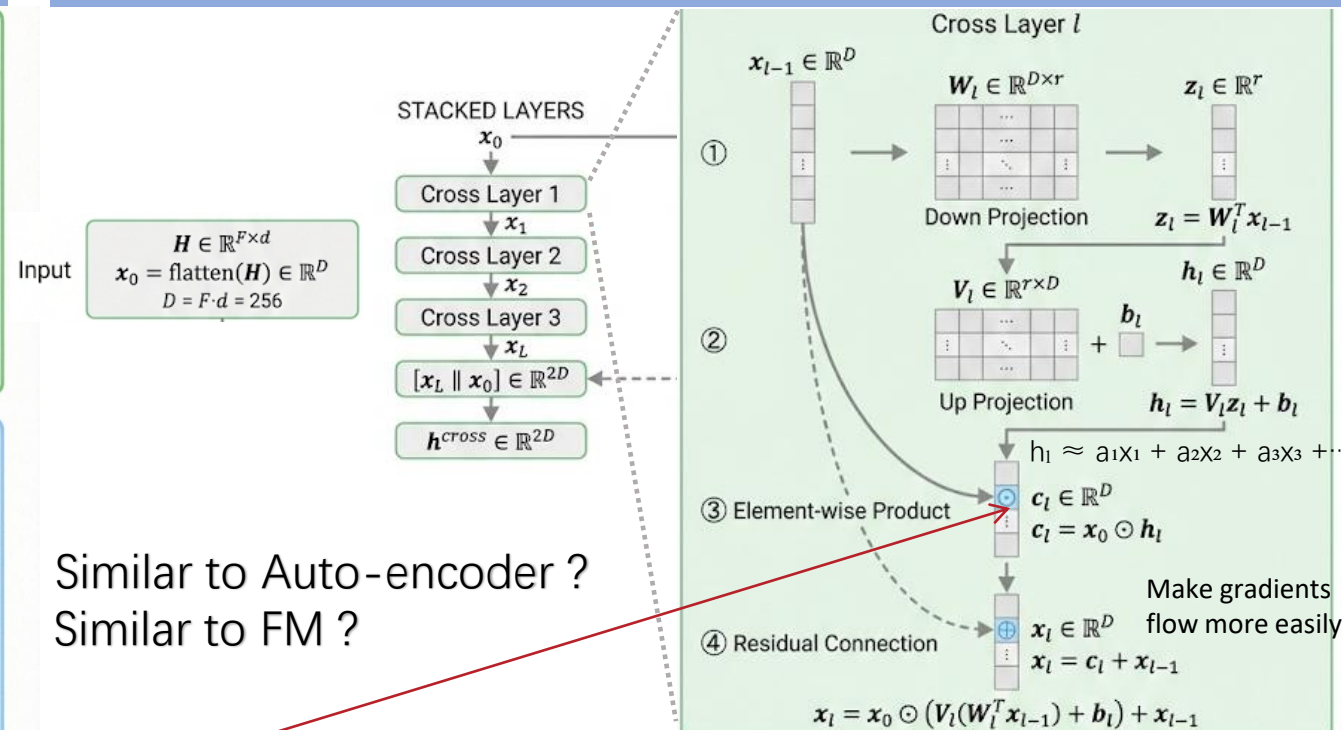
76.4



Stage 2-3: GBDT Ensemble & Feature Embedding



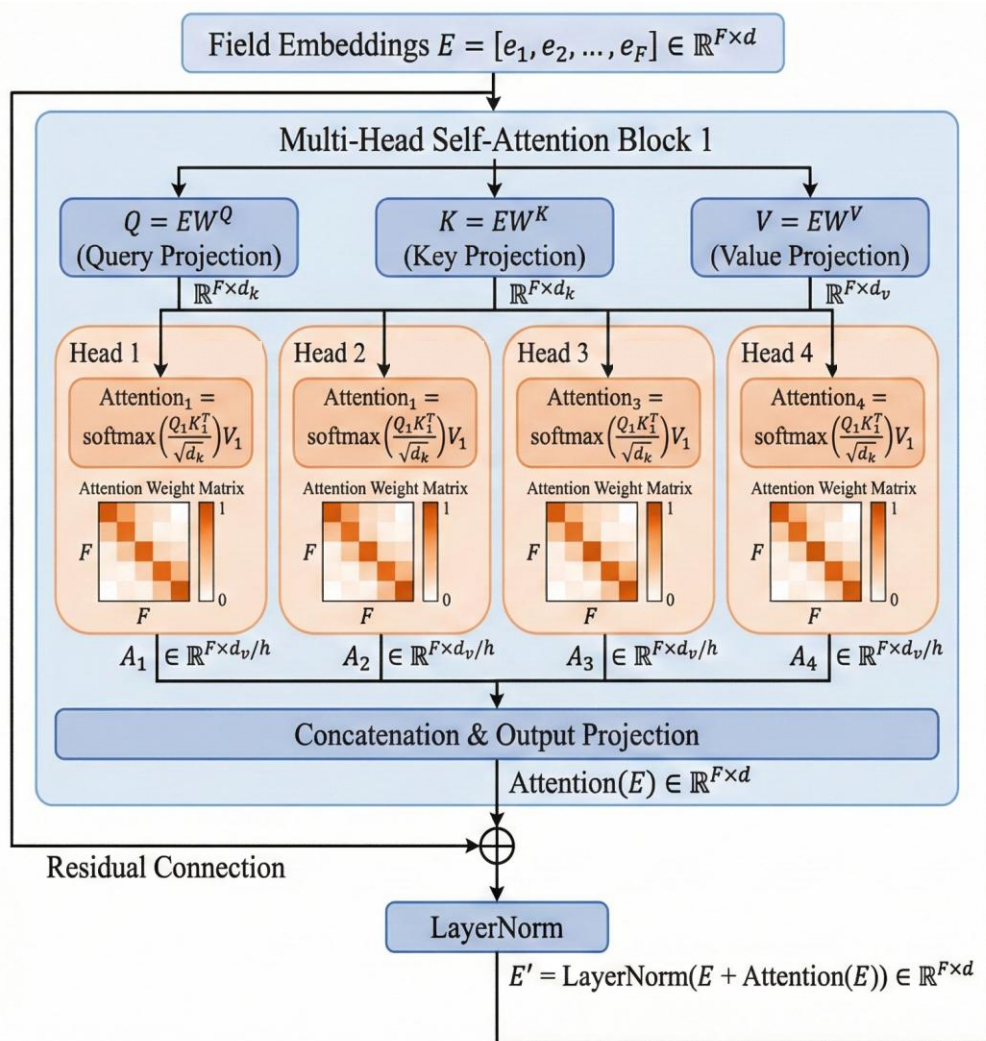
Stage 4 : DCN-V2 Deep Cross Network V2



Similar to Auto-encoder ?
Similar to FM ?

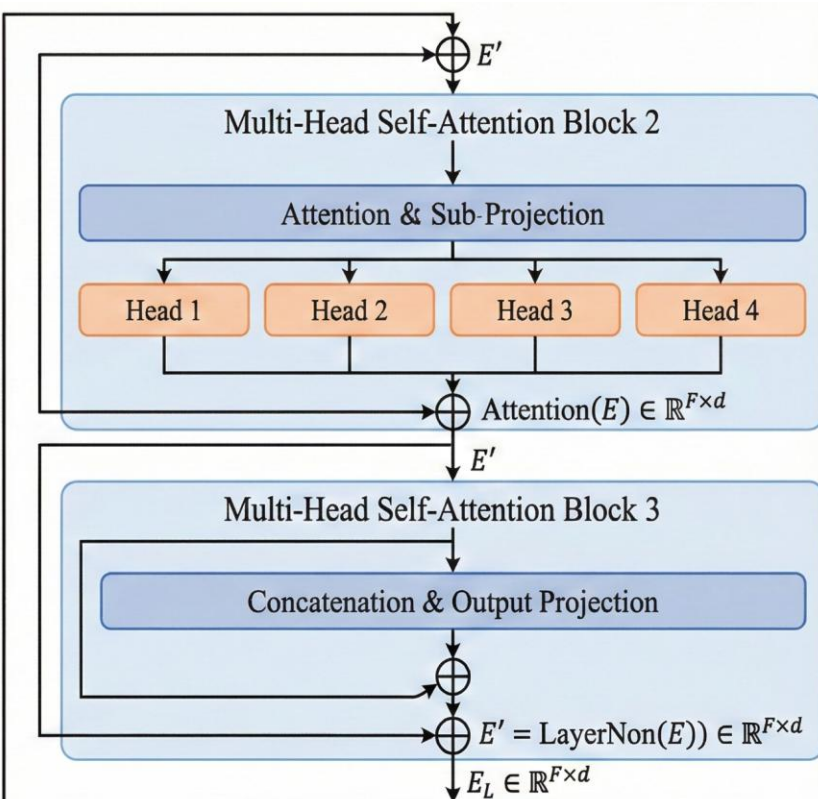
- ① $x_0 \odot h$ is the essence of feature crossing : A single operation can generate all second-order cross terms. (包括了所有交互项信息)
- ② Each additional layer increases the maximum interaction order by 1, but the number of parameters grows linearly. (虽然交互项最大阶数每层指数增加1, 但是模型参数是线性增加)
- ③ $D \rightarrow r \rightarrow D$: Use two small matrices ($D \times r$ and $r \times D$) instead of a large matrix ($D \times D$), reducing the number of parameters by about $D/(2r)$ times. (先降维后升维, 使得训练参数大大缩小)

Stage5 : Automatic Feature Interaction Learning via Self-Attentive Neural Networks



Through the attention weight matrix, you can directly see:

- which features have strong interactions 强交互
- which other feature that each feature pays attention to 特征间关联



Flatten $\rightarrow h^{attn} \in \mathbb{R}^{f \cdot d}$

Stage 5

= Transformer Self-Attention

+ 去掉位置编码 (特征顺序无所谓)

Eg: 狗咬人、人咬狗

{面积=100, 价格=500}

{价格=500, 面积=100}

+ 可以选择去掉 FFN 后面有 MLP

FFN 本质上就是:

逐位置单独做一次"升维→激活→降维"的非线性处理

+ 表格数据特化 (接入 MLP)

Output = Flatten(Attention_Output)

而不是:

Output = Attention_Output[0] #[CLS]

Output = Mean(Attention_Output)

$h^{cross} \in \mathbb{R}^{2D}$

DCN-v2

